Physics Laboratory Report: Motion on an Inclined Plane

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Abstract

The experiment was carried out by first calculating the surface's μ_s (*Static friction coefficient*) with the air compressor on when the steel air disc is just at rest. μ_k (*Kinetic friction coefficient*) was calculated after gathering the data, which is produced by measuring the ΔX for each interval generated by the spark oscillator that sparks every $0.02s$ throughout the experiment. The gathered data is used to calculate the acceleration and velocity quantities with the presence and absence of a pulley which is essential for calculating μ_k.

Introduction

The motion of objects on inclined planes adheres to the principles laid out by Newton's *Laws of Motion*.

$$\sum \vec{F} = m\vec{a}, \quad \vec{F}_{action} = -\vec{F}_{reaction}, \quad \sum \vec{F} = 0 \Rightarrow a = 0$$

For an inclined plane, additional factors have to be considered such as the angle θ of the incline and the force of gravity.

The angle of the incline is the angle formed between the inclined plane and the horizontal surface. As the incline becomes steeper, the force of gravity acting on the object also increases proportionally. The force of gravity is a force that attracts objects towards the center of the Earth and is always acting in a downward direction.

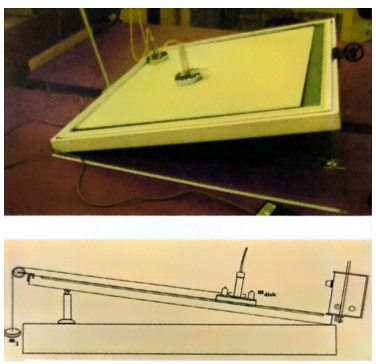
The experiment examines how a pulley system affects the acceleration of a steel air disc as it slides down an inclined plane. This was done by timing the disc's descent over a fixed distance with and without the pulley. Through analyzing the relationship between the pulley's presence, the incline angle, and

the recorded times, the goal was to understand the pulley's role in the disc's acceleration and its interaction.

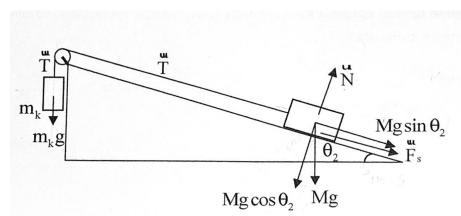
Procedure

The experiment setup is shown in Figure 1a. In this experiment, the air compressor is used to blow air underneath the steel air disc to lower the friction. The experiment was conducted with the air compressor in operation, allowing for the calculation of μ_s, μ_k and computing $\vec{v}$, $\vec{a}$ under these conditions. Experimentally, The friction coefficients were expected to be low, with the understanding that they could not reach zero due to physical constraints.

1. **Setup**: The inclined plane was fixed at an angle of θ degrees. The steel air disc was placed at the top of the inclined plane, and a pulley system was attached to the disc. An air compressor was used to reduce friction under the disc.



(a) Motion on an inclined plane experiment setup & setup diagram.



(b) The forces acting on the disc.

Figure 1: Setup.

2. **Measurement of μ_s**: The angle θ of the incline was adjusted until the disc was just at rest. This angle was recorded as the angle for which μ_s was measured. The static friction coefficient, μ_s, was calculated using the equations:

$$T = mgsin\theta + \mu_s N$$
$$T = mgsin\theta + \mu_s mgcos\theta$$
$$m_k g = mgsin\theta + \mu_s mgcos\theta$$
$$\mu_s = \left| \frac{m_k g - mg \sin\theta}{mg \cos\theta} \right|,$$

Where, m_k represents the mass of the hanged object, m is the mass of the disc, and g is the acceleration due to gravity. Using $m_k = 0.135$ kg, $m = 0.527$ kg, and $\theta = 17.7°$, we calculated μ_s to be 0.051.

3. **Data Collection**: Using a spark oscillator, we measured ΔX for every 0.02 seconds to calculate velocity and acceleration. The equations used were:

$$\vec{v}_n = \frac{\Delta X_n}{0.02} m/s \qquad \vec{a}_n = \frac{\Delta \vec{v}_n}{0.02} m/s^2$$
$$\vec{v}_{av} = \frac{\sum_{i=0}^{n} \Delta x_i}{n0.02} \qquad \vec{a}_{av} = \frac{\sum_{i=0}^{m} \vec{a}_i}{m}$$

4. **Calculation of μ_k:**

   For the pulley system, we used the equations:

   T from *free-body* digram on m_k

   $$m_k a = T - m_k g$$
   $$T = m_k a + m_k g$$

   Applying *free-body on m*

   $$\vec{F} = m\vec{a} = mgsin\theta - T - f_k$$
   $$m\vec{a} = mgsin\theta - T - \mu_k mgcos\theta$$

   Substituting T and solving for μ_k resulting the following:

   $$\mu_k = \left| \frac{ma - mgsin\theta + m_k(g + a)}{mgcos\theta} \right|$$

   For the non-pulley system, the equations were:

   $$ma = mgsin\theta - f_k$$
   $$\mu_k mg\cos\theta = mgsin\theta - ma \text{ (where } f_k = \mu_k mg\cos\theta)$$

   Solving for μ_k:

   $$\mu_k = \left| \frac{mg\sin\theta - ma}{mg\cos\theta} \right|.$$

5. **Data Analysis**: We gathered ΔX data for both systems and calculated μ_k values. Figures 2a and 2b show the data and variables for the pulley and non-pulley systems, respectively. These figures were used to compare μ_k values calculated from acceleration data for each system.

Results

The ΔX is measured to calculate $\vec{v}_{av}$, $\vec{a}_{av}$ and then compute μ_k of the system under two conditions: one employing a pulley system and another without one. In the setup with the pulley, the $\vec{v}_{av}$ was 0.425 m/s while $\vec{a}_{av} = 0.565$ m/s^2 which results with a μ_k value of 0.015. However, when we removed the pulley,

1	Time	Delta_x	Velocity	Acceleration
2	0.02	0.005	0.25	0
3	0.04	0.005	0.25	0
4	0.06	0.005	0.25	2.5
5	0.08	0.006	0.3	1.25
6	0.1	0.0065	0.325	0
7	0.12	0.0065	0.325	0
8	0.14	0.0065	0.325	1
9	0.16	0.0069	0.345	0
10	0.18	0.0069	0.345	0
11	0.2	0.0069	0.345	1.5
12	0.22	0.0075	0.375	0
13	0.24	0.0075	0.375	0.75
14	0.26	0.0078	0.39	0
15	0.28	0.0078	0.39	0.5
16	0.3	0.008	0.4	1.75
17	0.32	0.0087	0.435	0.25
18	0.34	0.0088	0.44	0.25
19	0.36	0.0089	0.445	0.5
20	0.38	0.0091	0.455	1
21	0.4	0.0095	0.475	0.5
22	0.42	0.0097	0.485	0
23	0.44	0.0097	0.485	0.5
24	0.46	0.0099	0.495	0
25	0.48	0.0099	0.495	0.25
26	0.5	0.01	0.5	1.25
27	0.52	0.0105	0.525	0.75
28	0.54	0.0108	0.54	0.25
29	0.56	0.0109	0.545	0.25
30	0.58	0.011	0.55	0
31	0.6	0.011	0.55	1.25
32	0.62	0.0115	0.575	1.25

(a) Data & variables with pulley system.

1	Time	Delta_x	Velocity	Acceleration
2	0.02	0.007	0.35	1.25
3	0.04	0.0075	0.375	2.5
4	0.06	0.0085	0.425	3.75
5	0.08	0.01	0.5	2.5
6	0.1	0.011	0.55	6.25
7	0.12	0.0135	0.675	1.25
8	0.14	0.014	0.7	2.5
9	0.16	0.015	0.75	2.5
10	0.18	0.016	0.8	0
11	0.2	0.016	0.8	7.5
12	0.22	0.019	0.95	2.25
13	0.24	0.0199	0.995	2.75
14	0.26	0.021	1.05	5
15	0.28	0.023	1.15	5
16	0.3	0.025	1.25	2.5
17	0.32	0.026	1.3	2.5
18	0.34	0.027	1.35	5
19	0.36	0.029	1.45	0

(b) Data & variables with non-pulley system.

Figure 2: Raw Data & variables.

$\vec{v}_a v$ increased to 0.888 m/s in this manner, $\vec{a}_{av}$ is also increased to 3.056 m/s^2 which Leads to μ_k being 0.032 as demonstrated in Figure 3.

Our previous measurement established a static friction coefficient μ_s of 0.051. Continuing the analysis, data visualization (refer to Figure 4) and the application of a regression model were used to illustrate the graphical findings.

1		Pulley system	Non-pulley system
2	Average Velocity (m/s)	0.425	0.888
3	Average Acceleration (m/s²)	0.565	3.056
4	Kinetic Friction Coeffitiont (μ_k)	0.015	0.032

Figure 3: Data Results.

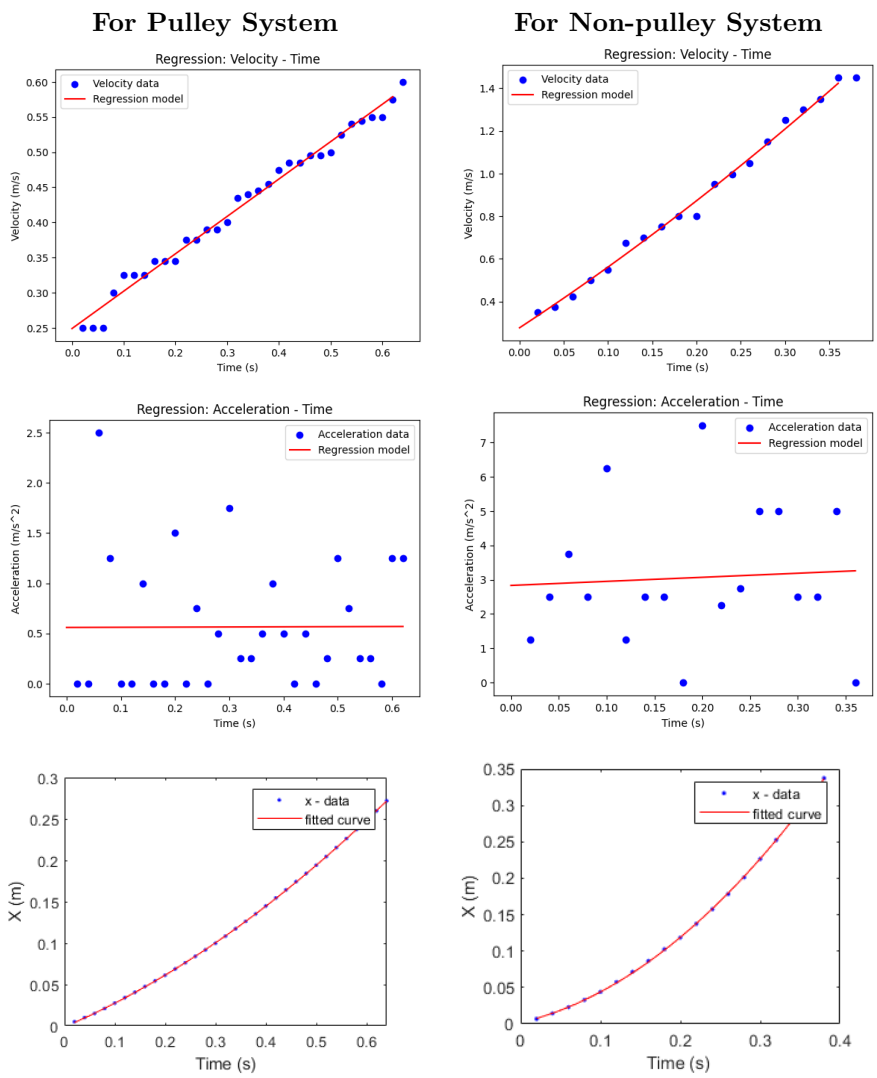


Figure 4: Data Visualization.

Analysis and Discussion

Having presented the results, the focus shifts to a detailed analysis and discussion of these findings.

Effect of Pulley System

The presence of a pulley system significantly affects the acceleration of the steel air disc. With the pulley, the average acceleration was found to be 0.565 m/s^2, while without the pulley, it increased to 3.056 m/s^2. This substantial difference demonstrates the pulley's role in reducing the acceleration of the disc, likely due to the pulley providing additional resistance and changing the magnitude of the force applied.

Friction Coefficients

The calculated kinetic friction coefficient (μ_k) also varies between the two setups. With the pulley, μ_k was calculated to be 0.015, whereas without the pulley, it increased to 0.032. This indicates that the presence of the pulley system reduces the friction between the disc and the inclined plane, allowing for smoother motion.

Comparison of Methods

The experimental approach of measuring ΔX for each time interval proved to be effective in calculating velocity and acceleration. This method provided accurate data for both the pulley and non-pulley systems, enabling the calculation of friction coefficients and the comparison of results. Visualization of the data, as shown in Figure 4, allows us to check that the models and data behave as expected for each graph plotted. This confirms the validity of the experimental setup and the reliability of the data obtained.

Limitations and Sources of Error

It is important to note that several factors could have introduced errors into the experiment. Air resistance, slight variations in the incline angle, and imperfections in the pulley system are potential sources of error that could have impacted the results. Additionally, the assumption of uniform acceleration may not hold true in real-world scenarios, which could have affected the accuracy of the calculated values.

Future Improvements

To improve the accuracy of the experiment, measures could be taken to minimize sources of error. Using a more precise incline angle measurement, reducing air resistance, and ensuring the pulley system operates smoothly could all contribute to more reliable results. Additionally, conducting the experiment multiple times and averaging the results could help mitigate random errors.

Conclusion

The experiment conducted to study the motion of a steel air disc on an inclined plane provided valuable insights into the effects of a pulley system on acceleration and friction coefficients. The presence of the pulley system was found to significantly reduce the acceleration of the disc, demonstrating the pulley's role in providing additional resistance. This effect was reflected in the calculated kinetic friction coefficients, which were lower with the pulley compared to without.

The comparison of methods, particularly the measurement of ΔX for each time interval, proved to be effective in calculating velocity and acceleration. The visualization of data through graphs allowed for a clear understanding of the relationship between variables and provided a means to verify that the models and data behaved as expected.

Limitations of the experiment included potential errors introduced by factors such as air resistance, variations in the incline angle, and imperfections in the pulley system. These limitations highlight the importance of careful experimental design and execution to minimize sources of error.

To improve the accuracy of future experiments, measures could be taken to reduce these sources of error, such as using more precise measurement tools and ensuring the experimental setup is free from external influences.

In conclusion, the experiment provided valuable insights into the motion of objects on inclined planes and demonstrated the importance of factors such as friction and pulley systems in determining the motion of objects.

Additional Resources

- Google Colab Notebook For Pulley System: Detailed notebook containing extra data, analysis, and calculations for this experiment.
  Pulley system:
  `https://colab.research.google.com/drive/1cnOkJYr3xsC9MOIJDzfbR63yPUdSOQ_X?usp=sharing`
  Non-pulley system:
  `https://colab.research.google.com/drive/1F2C_cCBMcDd6slDl_Ag2Df8XbSmyY9K4?usp=sharing`

- MATLAB Fit function documentation: This fitting technique is used specifically for ***X - Time*** graphs in Figure 4.
  `https://www.mathworks.com/help/curvefit/fit.html`